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designated departments to handle the programming and functioning of robotics.

According to Dr Venkatesh, "The focus is not on the production industry but on the research sector that is putting out amazing innovative applications involving advanced engineering techniques."

Dr Rout informs, "Companies like

Robotics is going to be one of the biggest industrial catalysts and major factors for growth in production and material handling facilities. Engineering students can avail this opportunity by selecting robotics as one of the career options

Texas Instruments (TI), Intel, Cypress, Infineon and STMicroelectronics, which are mostly into IC design, visit the BITS Pilani campus regularly."

Major international players in robotics like Energid Technologies, iRobot Corporation, Kuka Robotics Corporation, ABB Robotics, Fanuc and Motoman Robotics have their presence in India.

Pari Robotics, Robosoft Systems, ThinkLABS and Gridbots Technologies are some of the domestic players in the field of robotics.

Besides, several small and medium enterprises (SMEs) are also mulling over investing in robotics to increase their global competitiveness.

Companies look for different requirements while hiring a robotics engineer: Design and development capability in the domains of mechanical, manufacturing or electronics engineering, or as simple as operating and programming a specific robot for various needs.

Jayakrishnan explains, "Obviously, the requirements in this field necessitate a quality learning centered around the fundamentals of physics, dynamics and mathematics in general. Specific to our company, we would like engineers to have some hardware integration experience and, of course, good software coding skills and vision algorithm knowledge too. Another thing is making the software easy to use from a non-coder standpoint. So the ability to create a good user interface (human factors) to simplify the process is always in demand."

"As far as freshers are concerned, the amount of hands-on experience with robotic hardware and integration would depend on the student's desire as well as the quality of the institu-

tion. Apart from the knowledge acquired from the institution, a shorter learning curve with good grasping, strong mathematics and ability to conceptualise a solution are important. As India excels in software, we should be able to apply this excellence to robotics as it grows in global significance," he adds.

However, "At present, this professionally advanced industry is facing a severe lack of intelligent and skilled workers. It is the rampant state of the wrong people in the wrong place. The technical industry is handicapped due to the lack of relevant workforce and has to make-do with leftovers from the software industry," shares Dr Venkatesh.

"Companies look for practical, creative and innovative engineers who can integrate technology into real-life applications. This is just a dream as yet, as there is a lot of technology around and a lot less implementation," he adds.

From classroom to industry

Recruiters often have to spend their valuable time and resources in preparing freshers for specific jobs. This is usually because the candidates' knowledge of robotics is from a notional perspective only. So take up some industry-oriented project or internship. Various companies (in both public and private sectors) accommodate final- and pre-final-year students in their live projects.

At the same time, keep yourself updated with the latest technological trends to get an edge over your competitors. ●

The author is from EPF Bureau, New Delhi